

--- Performing LIME Analysis for Top Models ---

--- Generating LIME explanations for ExtraTreesClassifier ---

Explanation for: Correctly Classified Positive (Actual: 1, Predicted: 1) from ExtraTreesClassifier

Prediction probabilities

No Depression

Depression

No Depression 0.01

Depression 0.99

0.00 < What\_is\_your... 0.36  
 What\_is\_your\_CGP... 0.32  
 What\_is\_your\_CGP... 0.24  
 What\_is\_your\_CGP... 0.18  
 What\_is\_your\_CGP... 0.17  
 What\_is\_your\_CGP... 0.16  
 What\_is\_your\_CGP... 0.16  
 -0.65 < Do\_you\_ha... 0.14  
 What\_is\_your\_CGP... 0.13  
 What\_is\_your\_CGP... 0.12

Feature

Value

What\_is\_your\_CGPA\_3\_00\_3\_49 1.00  
 What\_is\_your\_CGPA\_3\_50\_4\_00 0.00  
 What\_is\_your\_CGPA\_3\_50\_3\_90 0.00  
 What\_is\_your\_CGPA\_3\_40\_3\_90 0.00  
 What\_is\_your\_CGPA\_3\_50\_3\_86 0.00  
 What\_is\_your\_CGPA\_3\_50\_4\_00 0.00  
 What\_is\_your\_CGPA\_3\_56\_3\_80 0.00  
 Do\_you\_have\_Anxiety\_ 1.53  
 What\_is\_your\_CGPA\_3\_00\_3\_50 0.00  
 What\_is\_your\_CGPA\_3\_20\_3\_89 0.00

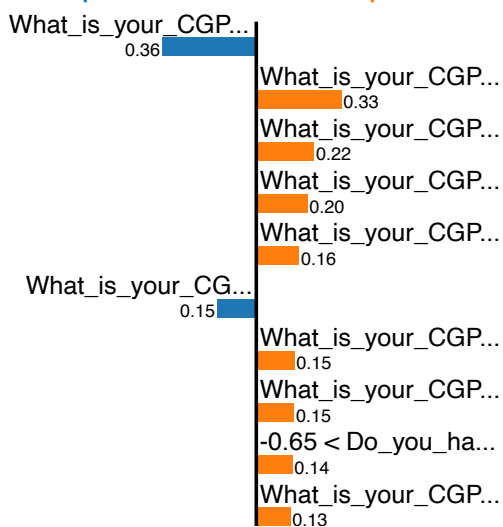
Explanation for: Correctly Classified Negative (Actual: 0, Predicted: 0) from ExtraTreesClassifier

## Prediction probabilities



## No Depression

## Depression



## Feature

## Value

|                             |      |
|-----------------------------|------|
| What_is_your_CGPA_3_00_3_49 | 0.00 |
| What_is_your_CGPA_3_50_4_00 | 0.00 |
| What_is_your_CGPA_3_50_3_90 | 0.00 |
| What_is_your_CGPA_3_40_3_90 | 0.00 |
| What_is_your_CGPA_3_50_3_86 | 0.00 |
| What_is_your_CGPA_3_20_3_89 | 1.00 |
| What_is_your_CGPA_3_56_3_80 | 0.00 |
| What_is_your_CGPA_3_50_4_00 | 0.00 |
| Do_you_have_Anxiety_        | 1.53 |
| What_is_your_CGPA_3_00_3_50 | 0.00 |

Explanation for: Misclassified (Actual: 0, Predicted: 1) from ExtraTreesClassifier

## Prediction probabilities



## No Depression

## Depression

What\_is\_your\_CGP...

0.36

What\_is\_your\_CGP...

0.33

What\_is\_your\_CGP...

0.24

What\_is\_your\_CGP...

0.20

What\_is\_your\_CGP...

0.17

What\_is\_your\_CGP...

0.15

-0.65 &lt; Do\_you\_ha...

0.15

What\_is\_your\_CGP...

0.14

What\_is\_your\_CGP...

0.14

What\_is\_your\_CGP...

0.13

## Feature

## Value

What\_is\_your\_CGPA\_3\_00\_3\_490.00

What\_is\_your\_CGPA\_3\_50\_4\_000.00

What\_is\_your\_CGPA\_3\_50\_3\_900.00

What\_is\_your\_CGPA\_3\_40\_3\_900.00

What\_is\_your\_CGPA\_3\_50\_3\_860.00

What\_is\_your\_CGPA\_3\_20\_3\_890.00

Do\_you\_have\_Anxiety\_ 1.53

What\_is\_your\_CGPA\_3\_56\_3\_800.00

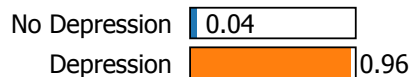
What\_is\_your\_CGPA\_3\_00\_3\_500.00

What\_is\_your\_CGPA\_3\_50\_4\_000.00

--- Generating LIME explanations for Random Forest ---

Explanation for: Correctly Classified Positive (Actual: 1, Predicted: 1) from Random Forest

## Prediction probabilities



No Depression

Depression

What\_is\_your\_CGP...

0.12



Feature

Value

|                             |      |
|-----------------------------|------|
| What_is_your_CGPA_3_00_3_49 | 1.00 |
| What_is_your_CGPA_3_50_4_00 | 0.00 |
| What_is_your_CGPA_3_50_3_90 | 0.00 |
| What_is_your_CGPA_3_56_3_80 | 0.00 |
| Do_you_have_Anxiety_        | 1.53 |
| What_is_your_CGPA_3_00_3_50 | 0.00 |
| What_is_your_CGPA_2_00_2_49 | 0.00 |
| What_is_your_CGPA_3_50_3_86 | 0.00 |
| What_is_your_CGPA_3_50_4_00 | 0.00 |
| What_is_your_CGPA_3_40_3_90 | 0.00 |

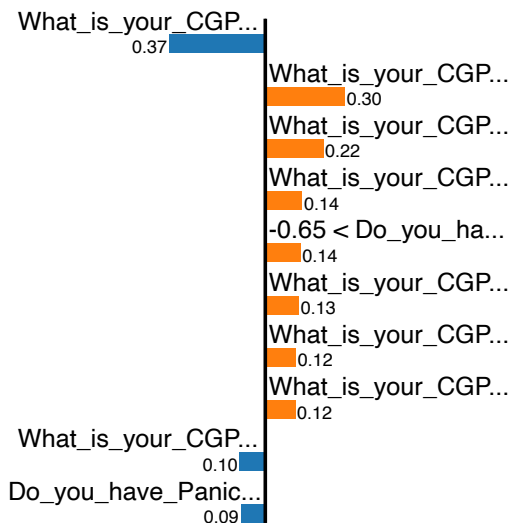
Explanation for: Correctly Classified Negative (Actual: 0, Predicted: 0) from Random Forest

# Prediction probabilities



## No Depression

## Depression



| Feature                     | Value |
|-----------------------------|-------|
| What_is_your_CGPA_3_00_3_49 | 0.00  |
| What_is_your_CGPA_3_50_4_00 | 0.00  |
| What_is_your_CGPA_3_50_3_90 | 0.00  |
| What_is_your_CGPA_3_56_3_80 | 0.00  |
| Do_you_have_Anxiety_        | 1.53  |
| What_is_your_CGPA_3_50_4_00 | 0.00  |
| What_is_your_CGPA_3_40_3_90 | 0.00  |
| What_is_your_CGPA_3_00_3_50 | 0.00  |
| What_is_your_CGPA_2_00_2_49 | 0.00  |
| Do_you_have_Panic_attack_   | -0.84 |

Explanation for: Misclassified (Actual: 0, Predicted: 1) from Random Forest

## Prediction probabilities

|               |      |
|---------------|------|
| No Depression | 0.46 |
| Depression    | 0.54 |

## No Depression

## Depression

What\_is\_your\_CGP...

0.36

What\_is\_your\_CGP...

0.30

What\_is\_your\_CGP...

0.23

-0.65 &lt; Do\_you\_ha...

0.14

What\_is\_your\_CGP...

0.13

What\_is\_your\_CGP...

0.13

What\_is\_your\_CGP...

0.12

What\_is\_your\_CGP...

0.11

What\_is\_your\_CGP...

0.11

What\_is\_your\_CGP...

0.10

## Feature

## Value

What\_is\_your\_CGPA\_3\_00\_3\_490.00

What\_is\_your\_CGPA\_3\_50\_4\_000.00

What\_is\_your\_CGPA\_3\_50\_3\_900.00

Do\_you\_have\_Anxiety\_ 1.53

What\_is\_your\_CGPA\_3\_56\_3\_800.00

What\_is\_your\_CGPA\_3\_00\_3\_500.00

What\_is\_your\_CGPA\_3\_40\_3\_900.00

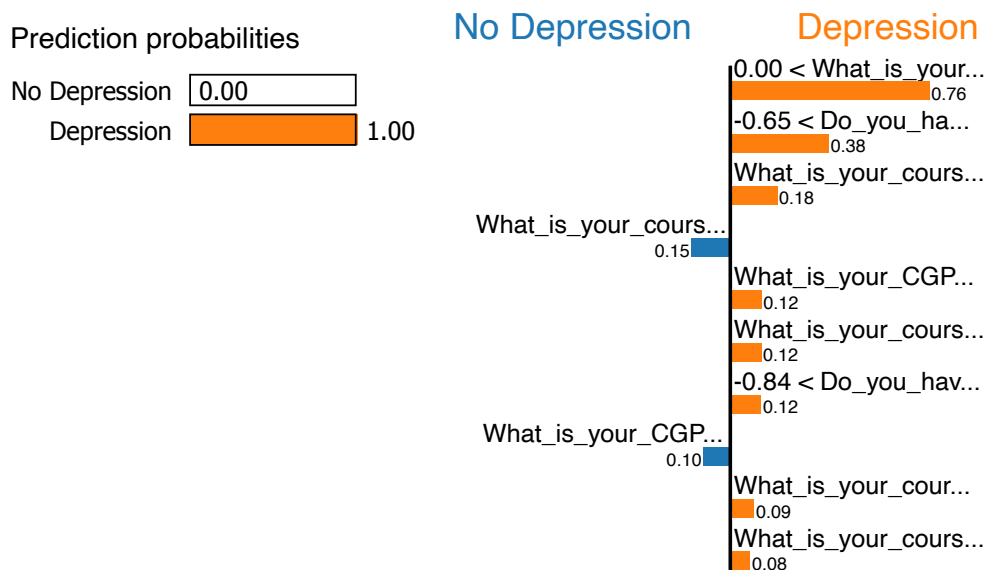
What\_is\_your\_CGPA\_3\_50\_3\_860.00

What\_is\_your\_CGPA\_3\_50\_4\_000.00

What\_is\_your\_CGPA\_2\_00\_2\_490.00

--- Generating LIME explanations for XGBoost ---

Explanation for: Correctly Classified Positive (Actual: 1, Predicted: 1) from XGBoost



| Feature                                             | Value |
|-----------------------------------------------------|-------|
| What_is_your_CGPA_3_00_3_49                         | 1.00  |
| Do_you_have_Anxiety_                                | 1.53  |
| What_is_your_course_Biotechnology                   | 0.00  |
| What_is_your_course_Molecular                       | 0.00  |
| What_is_your_CGPA_3_50_4_00                         | 0.00  |
| What_is_your_course_Persian_Language_and_Literature | 0.00  |
| Do_you_have_Panic_attack_                           | 1.19  |
| What_is_your_CGPA_3_30_3_60                         | 0.00  |
| What_is_your_course_Education_Research_Institute    | 0.00  |
| What_is_your_course_Film_Television                 | 0.00  |

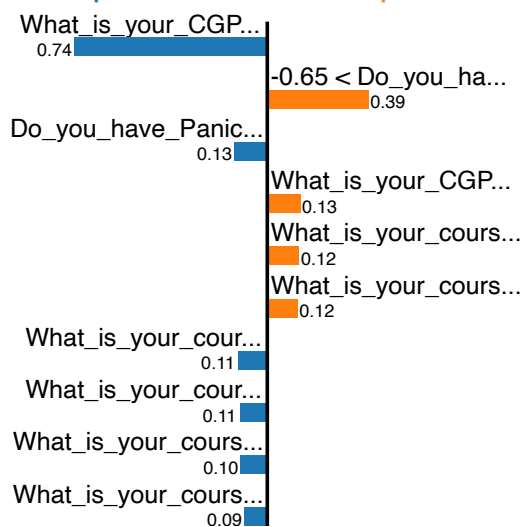
Explanation for: Correctly Classified Negative (Actual: 0, Predicted: 0) from XGBoost

## Prediction probabilities



## No Depression

## Depression

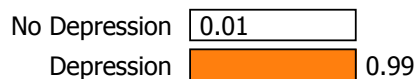


| Feature                                             | Value |
|-----------------------------------------------------|-------|
| What_is_your_CGPA_3_00_3_49                         | 0.00  |
| Do_you_have_Anxiety_                                | 1.53  |
| Do_you_have_Panic_attack_                           | -0.84 |
| What_is_your_CGPA_3_50_4_00                         | 0.00  |
| What_is_your_course_Pottery_Sculpture               | 0.00  |
| What_is_your_course_Biotechnology                   | 0.00  |
| What_is_your_course_Painting_Oriental_Art_Printing  | 0.00  |
| What_is_your_course_Education_Research_Institute    | 0.00  |
| What_is_your_course_Persian_Language_and_Literature | 0.00  |
| What_is_your_course_Zoology                         | 0.00  |

Explanation for: Misclassified (Actual: 0, Predicted: 1) from XGBoost

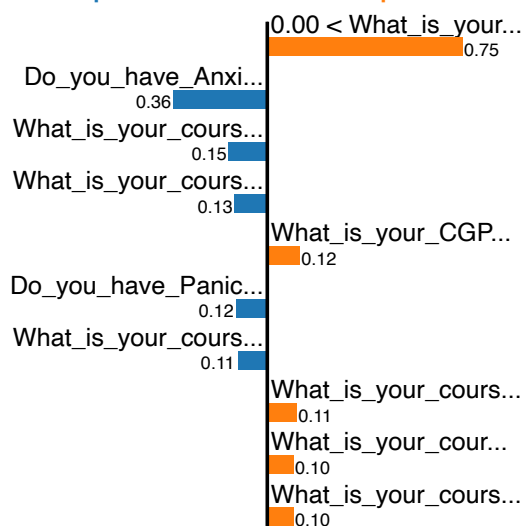


## Prediction probabilities



## No Depression

## Depression



## Feature

## Value

|                                                            |       |
|------------------------------------------------------------|-------|
| What_is_your_CGPA_3_00_3_49                                | 1.00  |
| Do_you_have_Anxiety_                                       | -0.65 |
| What_is_your_course_Robotics_and_Mechatronics_Engineering  | 0.00  |
| What_is_your_course_Land_Management_Law                    | 0.00  |
| What_is_your_CGPA_3_50_4_00                                | 0.00  |
| Do_you_have_Panic_attack_                                  | -0.84 |
| What_is_your_course_Genetic_Engineering                    | 0.00  |
| What_is_your_course_Persian_Language_and_Literature        | 0.00  |
| What_is_your_course_Operations_and_Supply_Chain_Management | 0.00  |
| What_is_your_course_Geography                              | 0.00  |

LIME analysis complete for selected models.